

ALX Professional Foundations: Milestone 3

Worksheet

Instructions: Provide responses to all items in the orange boxes. The worksheet consists of Sections A-E. Work on this worksheet one section at a time throughout your week, and return to Savanna after each section for the next set of content and further instructions.

SECTION A: Problem Statement

Step 1: Your GCGO

Which Grand Challenge or Great Opportunity (GCGO) do you want to play a part in addressing? (Pick one.)

As a reminder, the GCGOs are:

- Urbanization
- Education
- Infrastructure
- Healthcare
- Climate change
- Governance
- Job creation
- Agriculture
- Natural resources
- Arts, culture, and design
- Tourism
- Empowerment of women
- Regional integration
- Wildlife conservation

GCGO: healthcare

Step 2: Describe Your Problem

You are going to take a first pass at briefly describing your chosen problem. This can be any problem that speaks to you, as long as it is a real-life occurrence that is clearly linked to your chosen GCGO, that occurs in a certain place and for certain people (and/or animals), and that it can be clearly defined.

For example, if you chose wildlife conservation as your GCGO, you might first state your chosen problem as:

There are very few white rhinos left in Kenya and they are in danger of going extinct.

Another example, if you choose infrastructure as your GCGO:

Residents of major cities in South Africa endure prolonged periods without electricity, significantly hampering their ability to generate income.

Note that this is just your first attempt stating the problem, and you don't need to quantify the problem yet. In order to get to your official problem statement (which does need to be quantifiable) first answer the following questions. The more specific your answers, the better. You may also ask Google, Wikipedia, , ChatGPT, and/or other reliable online sources to help you. Please be sure to cite (give credit to) any sources that you use.

Describe your problem using What/Who/When/Where/Why/How....

1. **What** is the problem? What is reality like because of this problem? What will reality be like if the problem continues?

Healthcare in Africa faces challenges due to inadequate infrastructure, medical shortages, and limited funding, contributing to high rates of diseases and poor health conditions.

In several African countries, the availability of healthcare facilities ranges widely, from 0.1 to 3.5 facilities per 10,000 people. In many regions, over 50% of these facilities lack essential amenities such as electricity and clean water. The ratio of doctors per 10,000 population also varies significantly across these nations, ranging from 0.1 to 5 doctors per 10,000 people. Additionally, certain countries face significant healthcare workforce challenges, with vacancy rates for healthcare positions soaring as high as 30%, largely due to shortages or emigration of medical professionals.

Continued healthcare challenges in African countries may lead to exacerbated health disparities and compromised public health outcomes. Persistently high vacancy rates in

healthcare positions could strain remaining staff and reduce overall healthcare quality. Addressing these issues necessitates significant investment in infrastructure and workforce development to improve access and service delivery.

2. **Who** does this problem impact, directly and indirectly? Who contributes to the problem?

Millions across Africa are directly affected by limited access to healthcares. Contributing factors include inadequate investment in healthcare infrastructure, insufficient healthcare workforce planning, and social injustice in resource allocation.

3. **When** did this problem begin? When does it occur?

The healthcare challenges in Africa have started since colonial times, as it was originally shaped by systems with limited local focus. Today, issues like inadequate infrastructure, medical staff shortages, and issues in healthcare access and funding continue to impact healthcare delivery and outcomes all over the continent.

4. **Where** is this problem occurring? What is the context in which it occurs?

This problem is occurring throughout Africa, affecting both urban and rural areas across various countries on the continent. It occurs within the context of varying economic conditions, political environment, and levels of healthcare infrastructure and resources.

5. **Why** is this a problem? What are the pain points or gaps? Why do you personally care about this problem?

Healthcare is crucial because it impacts people's lives and the future of nations.

There is a need for adequate medical facilities, supplies, and medications to meet the needs of the population. Additionally, raising awareness about the importance of healthcare is essential through sufficient medical staff and equitable distribution of healthcare resources.

I come from a background where healthcare was not available to everyone at the needed standard, and where many people are not educated enough about the importance of keeping their bodies and minds healthy. I truly believe that if all people in Africa had a healthy life, our nations would prosper and develop in an exponential pattern.

6. **How** would reality be different if this problem were solved? (This can be your opinion.)

The very first impact to the advancement of healthcare will be the increase in IQ and cognitive abilities; A healthy mind lies in a healthy body. In addition, more focus will be directed to other areas like innovative development and economic growth, with the available healthy productive workforce that will implement all of the brilliant ideas to come.

Step 3: Understand and Quantify Your Problem

Next, you will conduct some basic web research to better understand, define, and quantify your problem. You will do this through a combination of Google search, Wikipedia, credible web sources, ChatGPT or other AI research tool, and your own synthesis of information from these sources. Be sure to give credit to your sources, and paraphrase (use your own words) rather than quoting directly.

7. What is the historical context for this problem? What happened in the past that contributes to the problem now?

According to chatGPT,

- Colonial influence established healthcare systems to prioritize colonial interests.
- Post-independence, Africa faced challenges inheriting underdeveloped healthcare infrastructure.
- Ongoing global economic disparities have limited Africa's healthcare investment capabilities.

8. What are the possible economic (money-related) reasons why this problem exists and continues?

- Poor economic systems in developing countries mean less investment in healthcare
- old and deteriorated healthcare systems cannot cope with the increased population in people with chronic diseases
- inadequate distribution of medical supplies and resources globally and limitation on the availability makes healthcare not affordable for many

9. What are the possible political reasons why this problem exists and continues?

- The continuing political conflicts in the region takes the attention of investment and spending away from the healthcare

- lack of efficient regional integration between neighboring countries to exchange resources, knowledge and experiences

10. What cultural beliefs and/or social norms possibly contribute to this problem?

- Cultural beliefs may influence preferences for traditional medicine over modern healthcare practices.
- Social norms could affect healthcare-seeking behaviors, particularly regarding gender roles or stigma related to certain health conditions.

11. Who are the people potentially responsible (directly or indirectly) for creating and/or maintaining this problem?

- Government officials and policymakers who allocate healthcare budgets and set healthcare policies.
- Healthcare administrators and providers responsible for managing healthcare facilities and services.
- International organizations and donors influencing healthcare funding and development priorities.
- Local community leaders and influencers shaping healthcare-seeking behaviors and cultural attitudes towards modern healthcare practices.

Now that you have the preliminary information you need, you'll continue your web research to find some numbers, or quantifiable information, to help describe your problem:

What numerical data can you find that is relevant to your problem? Be sure to use your own words and also cite (give credit to) your sources.

Example 1:

According to Chat GPT, there are about 880 white rhinos currently living in Kenya. This population is very small, and they are critically endangered.

Example 2:

Johannesburg has approximately 5.8 million residents (per ChatGPT) and had approximately 4.7 million international overnight visitors in 2019 (according to the South African Tourism Annual Report for 2019/2020).

12. Approximately how many people (and/or animals) are **directly** impacted by this problem? Explain.

According to Chat GPT, Approximately 1.3 billion people in Africa are directly impacted by challenges in healthcare access, including issues such as inadequate infrastructure, medical staff shortages, and disparities in healthcare provision.

13. Approximately how many people (and/or animals) are **indirectly** impacted by this problem? Explain.

Indirectly, the healthcare access problem in Africa affects whole communities and economies. This includes families caring for the sick, businesses facing productivity losses, and communities struggling with healthcare expenses. Zoonotic diseases further complicate matters, affecting both human health and agricultural productivity. Although precise figures are difficult to measure, these indirect effects touch millions across diverse sectors within African societies.

14. What other numerical data can you share that is relevant to your problem? What can you find out about its size and scope? What can be measured? (For example, the amount of trash produced in Nairobi each day, the number of people without access to clean water, etc.)

According to Chat GPT,

- In some African countries, the number of healthcare facilities ranges from 0.1 to 3.5 facilities per 10,000 people.
- Around 50% or more of healthcare facilities in certain regions lack basic amenities like electricity and clean water.
- The ratio of doctors per 10,000 population varies widely across African countries, ranging from 0.1 to 5 doctors per 10,000 people.
- Some countries experience a vacancy rate of healthcare positions as high as 30% due to shortages or emigration.

Step 4: Describe Your Solved State

Without having to come up with *how* to solve the problem, describe what the desired, solved state looks like. Please use numbers wherever possible, and make your solved state specific and measurable.

Example 1:

There would be a population of 10,000 healthy and protected white rhinos living in the wild in Kenya.

Example 2:

All 5.8 million residents of Johannesburg would have affordable and consistently available power from clean energy sources, 99.5% of the time.

15. If the problem were addressed/solved, what would reality be like?

- There would be equitable access to quality healthcare services for all people across the continent.
- Health outcomes would improve significantly, leading to reduced mortality rates, particularly among mothers and infants.
- Economic productivity would increase as healthier populations contribute more effectively to their communities and economies.
- Healthcare systems would be robust, with adequate infrastructure, well-trained medical professionals, and sustainable funding mechanisms in place to ensure long-term health and well-being.

16. Are there other benefits that would come from your problem being solved? Name at least one.

The very first impact to the advancement of healthcare will be the increase in IQ and cognitive abilities; A healthy mind lies in a healthy body. In addition, more focus will be directed to other areas like innovative development and economic growth, with the available healthy workforce that will implement all of the brilliant ideas to come.

Step 5: Clarify Your Problem Scope

You are more effective at solving a problem when you know where its limits are. That is, when you know what is “in scope” and “out of scope.” For this reason, it is important to list what is out of scope, or NOT included as part of your problem definition.

Example 1:

The scope of the problem does not cover any other animal species besides white rhinos. It does not include white rhinos outside of Kenya.

Example 2:

The scope of the problem does not include any businesses or people outside of the legally-defined Johannesburg city limits. It does not apply to tourists or visitors staying for less than 1 year in Johannesburg.

17. What is NOT in scope for your problem?

The scope of the healthcare problem doesn't include any area outside africa limits.

Step 6: Areas for Learning

What do you not know or understand that you would like to know more about? This can be anything related directly or indirectly to your problem. Let your curiosity run wild!

Example 1:

I'd like to know where most of the demand for rhino horn is coming from. I'd like to know who are the primary buyers and who is behind the trafficking of rhino horn. I'd like to know how long the average rhino's lifespan is. I'd like to know how many babies a typical female rhino has, and how many babies typically survive into adulthood. I'd like to know more about what diseases impact rhinos. I'd like to know more about the kinds of habitats that rhinos thrive in. I'd like to understand what international organizations do the best job supporting wildlife conservation and what their practices are. I'd like to know what models of community involvement have been most successful in keeping wildlife safe and thriving. I'd like to understand how much land is available in Kenya for rhinos to roam.

Example 2:

I'd like to better understand the utility company Eskom and its history. I'd like to understand why Eskom has failed to plan properly to update its infrastructure. I'd like to understand the relationship between Eskom and the South African government. I'd like to know if there are private utility companies providing competition to Eskom. I'd like to know what the latest breakthroughs are in solar power. I'd like to know what other possible energy sources might be made available in Johannesburg. I'd like to know how much energy tourists and temporary visitors use. I'd like to better understand the process of how limited energy supply gets allocated to people and businesses. I'd like to better understand the economic impact to people and businesses of not having power.

18. What else would you like to know or understand better? (It can be anything related to your problem.) List 5-10 things.

I want to know more about the most prevalent diseases in Africa, their causes of spread, the highest mortality rate due to inadequate healthcare in which country and year, the most affected countries during the COVID-19 pandemic, the most developed healthcare systems in Africa, and the top medical colleges in Africa.

Step 7: Problem Statement

This step is the culmination of all you have done in Part A. You will synthesize the work you have done above to create a problem statement of 150 - 250 words. This should be in narrative form, 2-4 paragraphs, and should NOT use bullet points.

Your problem statement should:

- Provide a succinct description of the problem **in the first sentence**.
- Indicate specific population affected
- Explain the impact (cost, time, environmental, personal) and why the problem matters.
- Explain what reality would look like if the problem were solved. The gap that exists between present reality and the desired outcome should be clear.

Please cite (give credit to) where your information came from directly in your statement. Avoid word-for-word quoting and instead paraphrase (use your own words), as modeled in the example. Also list your sources and their urls (web addresses) at the end.

Example :

Kenya's white rhinos are in critical danger of extinction. There are currently about 880 white rhinos in the country of Kenya, per Wikipedia. According to Chat GPT, Rhinos are considered a keystone species, meaning they have a disproportionately large impact on their ecosystem compared to their population size. Rhinos help shape their environment by influencing vegetation growth and acting as seed dispersers, which creates habitat for other species (per ChatGPT).

The extinction of white rhinos would have cascading effects on other plant and animal species in their habitat. According to the Kenya Wildlife Service, rhinos' presence in reserves and parks bring millions of tourists each year, contributing to local economies and supporting conservation efforts. Once a species goes extinct, it is gone forever. The extinction of rhinos would represent the loss of millions of years of evolutionary history, and unique genetic diversity that science has yet to fully understand and benefit from (per ChatGPT).

My problem would be considered solved when the population of wild, white rhinos in Kenya reaches 10,000, and when all imminent threats to their population including poaching and habitat destruction are not present. If this were the reality, it would create ecosystem balance, create large revenues from ecotourism, preserve important cultural symbols, and allow for genetic diversity that could benefit humanity in ways we may not yet fully understand.

Sources:

Kenya Wildlife Services Annual Report 2017,

<https://www.kws.go.ke/content/annual-reports>

ChatGPT, <https://chat.openai.com/>

"White Rhinoceros", Wikipedia, https://en.wikipedia.org/wiki/White_rhinoceros

19. My problem is statement is:

Access to adequate healthcare across Africa is severely limited due to lack of investment in healthcare infrastructure, insufficient healthcare workforce, and social injustice in resource allocation. These factors contribute to high rates of preventable diseases, maternal and infant mortality, and overall health issues among the

population. According to Chat GPT, Approximately 1.3 billion people in Africa are directly affected by limited access to healthcare, located in both urban and rural areas across various countries across the continent.

Let aside the economically worsened situation that is a result of unhealthy workforce, healthcare is crucial because it impacts people's lives and the future of nations. There is a need for adequate medical facilities, supplies, and medications to meet the needs of the population and the economic system in general.

I come from a background where healthcare was not available to everyone at the needed standard, and where many people are not educated enough about the importance of keeping their bodies and minds healthy. I truly believe that if all people in Africa had a healthy life, our nations would prosper and develop in an exponential pattern.

The advancement of healthcare will increase IQ and cognitive abilities of young individuals. Additionally, with this advancement, the awareness about the importance of modern healthcare practices will replace non-scientific cultural beliefs and traditional medicine. Finally, more governmental focus will be directed to other priorities like innovative development and economic growth, using available healthy and productive workforce that will implement the brilliant ideas to come.

20. Please list all sources that you used to form your problem statement:

resources:

- [World Health Day: 8 trends shaping global healthcare | World Economic Forum \(weforum.org\)](https://www.weforum.org/agenda/2020/04/world-health-day-8-trends-shaping-global-healthcare/)
- [ChatGPT](#)



Please go back to Savanna and continue with your learning content. You will be filling out Step 8 after you've completed the Peer Activity.

Step 8: Peer Activity Report

This step is related to your peer activity and should help you to improve your problem statement. Answer the following questions as part of the activity and then update your Problem Statement in Step 7 above, based on your feedback.

21. Who reviewed your problem statement? (Give the first and last names of your 2 peers).

Zinzile Mhlanga

Mitchell Ngarandi

22. In brief, what feedback did they give to you?

Zinzile Mhlanga feedback:

Loved your Milestone 3. I loved that you provides statistics to really show how the healthcare challenge has affected a lot of African countries.

Overall you did thorough research and keep up the good work!

Mitchell Ngarandi feedback:

She Gave me some comments to modify and gave me this feedback "Overall well done Christina. Your GCGO is easy to understand and well thought of."

23. Was their feedback useful to you? Did it feel kind? Why or why not?

Yes, the two feedbacks were very useful and kind to me.

They helped me see what I did well in writing my problem statement, what to modify and gave me positive feelings as well.

24. Who did you give feedback to? (Give the first and last names of 2 peers- they may be the same or different peers from above.)

Zinzile Mhlanga

25. Do you feel that you gave useful and kind feedback to your peers? Why or why not?

Yes, I think so because I highlighted the good things she did and encouraged her to keep up the good work.



Please go back to Savanna and continue with your learning content. You will be prompted on when to return to complete Section B.

SECTION B: Research Questions & Hypothesis

IMPORTANT: Complete this section AFTER completing the Savanna Modules *Asking Effective Questions* and *Web Research*.

Step 9: Research Questions

Based on what you have learned so far and on 'Step 6: Areas for Learning' from this worksheet, come up with 3 research questions. **Research questions should be complex enough that they can't be answered by a single Google search.** If appropriate, form a hypothesis that your research may confirm or reject. (As a reminder, a hypothesis is a

prediction of how you think your research will answer your research question. It is your best guess. If you truly have absolutely no idea, state "not applicable.")

Example Research Question #1:

What are some ways can we increase rhino populations?

Hypothesis:

Rhino populations will be increased by creating more open spaces for them to roam, increasing their protection, increasing international interest in them, and other reasons I have yet to uncover.

Example Research Question #2:

Which organizations have been effective at wildlife conservation and what practices do they use?

Hypothesis:

Not applicable; I don't know.

Example Research Question #3:

How many babies can a typical female white rhino have in her lifetime, and what are the reasons a female may not have high fertility?

Hypothesis:

A typical female white rhino can have 5 babies in her lifetime, and fertility may be affected by diet, amount of grazing territory, poaching, stress, mate availability, and other reasons I have yet to uncover.

26. Research question #1:

Which African countries were the most affected by COVID 19 and what actions can be taken to have better readiness against similar pandemics?

Hypothesis (if applicable):

Not applicable

27. Research question #2:

What are examples of successful stories about healthcare improvement in African countries and how can we copy the success stories to other African countries to spread this success?

Hypothesis (if applicable):

Applying the same approach of Hepatitis C virus campaign in Egypt can limit the spread of preventable diseases in other African countries

28. Research question #3:

What are the Healthcare system areas which need the most attention in the African countries?

Hypothesis (if applicable):

The Healthcare Infrastructure to be improved as a priority.



Please go back to Savanna and continue with your learning content. You will be prompted on when to return to complete Section C.

SECTION C: Web Research

Part 1: Research Plan

You can plan out your research, for each of your research questions, using the template below.

29. Step # 1: Define your objectives, 3 research questions & associated hypotheses.

Key Questions: What are you trying to accomplish with this research? What do you wish to find out that will accelerate your work in the right direction?

Your Response:

- 1- Data on the impact of COVID 19 on the various African countries
- 2- information about response plan of the least affected countries globally
- 3- ways to transfer the best practices to other countries

- 1-A healthcare problem that was overcome by an African country.
- 2-what they did to overcome this problem so that we can copy their successful stories.

- 1-some information about healthcare different sectors.
- 2-which sector most needed attention from us.

30. Step # 2: Determine your end outputs.

Key Questions: What type of data/information is ideal for you to find, based on what you are trying to accomplish? Do you need to present it in any particular format?

Your Response:

Text-based and visualized information, i can present it in a clear and structured manner in textual format.

31. Step # 3: Scope your main sources of information.

Key Questions: Do you already know where you might want to go to find the information you're seeking? Are there particular entities or organizations that you know are seen as "experts" on the topic?

Your Response:

Yes, I will check in google search and chat GPT, i think i will also check in World Health Organization (WHO) as an expert in this topic.



After you updated earlier parts of the worksheet based on your research plan, please go back to Savanna and continue with your learning content. Return to Part 2 and Part 3 prior to submitting your milestone.

Part 2: Conducting Research

You can answer the following questions as you complete conducting your research into your research questions.

32. Step 1: Begin gathering your required information.

Key Questions: What search terms should you use? How many reports/articles do you want to read before deciding you have seen enough?

Your Response:

I will use keyword Search Terms, Exact Phrase Search by using quotes and Field Search
I will decide that's enough after reading about 7-10 articles.

33. Step 2: Evaluate the Validity/Credibility of Your Sources and Information

Key Questions: Are the sources credible and reliable? Consider the authority, accuracy, objectivity, and currency of the information to ensure its validity for your research.

Your Response:

Yes, sure.

34. Step 3: Synthesize and Communicate Your Key Findings.

Key Questions: Summarize the key information and findings that you have gathered during your research. Organize these findings in a clear and coherent manner, ensuring that they directly address your research objectives and questions.

Your Response:

Yes, done

35. Return to Your Original Objectives and Key Questions.

Key Questions: Revisit your initial objectives and key questions to ensure that your research findings adequately address them. Reflect on whether your findings have effectively contributed to solving the identified problem.

Your Response:

Yes, done.

Part 3: Research Summary

In 200-300 words, provide an executive summary of your research in the text box below. You should be synthesizing information from multiple sources. Provide answers and explanations for the 3 questions you investigated and your key research findings. This should be in a narrative format (no bullet points), and be at least 3 paragraphs long.

Please use at least 3 different online sources such as ChatGPT, organizational websites, Wikipedia, etc. Please cite (give credit to) where your information came from directly in your statement. Avoid word-for-word quoting, paraphrase instead (use your own words), as modeled in the example. Also, list your sources and their URLs (web addresses) at the end.

36. Research Summary

summary:

In mid-2020, Africa reported 251,866 COVID-19 cases, with 6,769 deaths and 114,308 recoveries across 55 countries, comprising 3.2% of global cases. South Africa, Egypt, Nigeria, Ghana, and Cameroon accounted for 63% of these cases, revealing significant disparities influenced by healthcare infrastructure, population density, and socio-economic factors. Addressing these disparities and enhancing readiness against future pandemics requires bolstering healthcare systems, improving surveillance, and investing in public health

education, international collaboration, and tailored vaccines and treatments. However, national responses often overlooked local needs, hindering community engagement and pandemic preparedness.

Successful healthcare improvements in Rwanda (community health insurance), Botswana (HIV/AIDS response), and Ethiopia (Health Extension Program) demonstrate the importance of political commitment, community involvement, and partnerships with international organizations. To replicate these achievements continent-wide, prioritizing political leadership in healthcare, engaging communities in decision-making, and adapting interventions to local contexts are crucial. Investments in healthcare infrastructure, workforce capacity, and health information systems are essential to improve access and quality of healthcare services across Africa.

Critical areas requiring immediate attention include inadequate healthcare facilities and equipment in rural areas, shortages of trained healthcare professionals, financial constraints impacting medication availability, and weak health information systems hindering disease monitoring and resource allocation. Strengthening these areas through comprehensive strategies will enhance healthcare delivery, resilience against pandemics, and ensure equitable access to healthcare for all African populations.

Resources:

- [COVID-19 prevalence estimation: Four most affected African countries - PMC \(nih.gov\)](#)
- [COVID-19 in Africa: rethinking the tools to manage future pandemics - PMC \(nih.gov\)](#)
- [ChatGPT](#)
- [Strengthening the basics: public health responses to prevent the next pandemic | The BMJ](#)
- [How COVID gave African countries the opportunity to improve public health \(theconversation.com\)](#)
- [Rwanda's Health Care Success Story - The New York Times \(nytimes.com\)](#)
- [Acting now to strengthen Africa's health systems \(mckinsey.com\)](#)
- [Rwanda's primary health care strategy improves access to essential and life-saving health services \(who.int\)](#)

Once you have completed this worksheet, export/convert to .pdf, rename it per the instructions, and upload to Savanna as your Milestone # 3 Submission. Celebrate a job well done!